

Antiwork Topic Modeling

Exploratory longitudinal analysis for collaborator review

97,254	199	38.6%	19.0%
filtered posts	non-outlier clusters	topic assignment	theme-map coverage

Study summary

Across 97,254 management-related r/antiwork posts from March 2021 through February 2025, the updated BERTopic analysis identifies a descriptive shift away from COVID-centered health and safety discussion and toward scheduling and time-off discussion. Compensation, career precarity, and recruitment remain present across the four rolling windows.

Completed analysis

Rebuilt the analysis with BERTopic; exported document-level assignments and four rolling-window fits; finalized a 27-cluster, nine-theme codebook; and produced figures and reproducibility materials.

Interpretive scope

The figures show the share of all filtered posts captured by included candidate-theme clusters. The 27-cluster map is not a full taxonomy or a complete estimate of management-related content.

Selected findings

Theme	Y1	Y4	Interpretation
Health safety and attendance	4.97%	2.50%	Down 2.47 pp; early COVID/illness concentration contracts.
Scheduling hours and time off	1.71%	3.20%	Up 1.49 pp; largest mapped theme in Y4.
Compensation and pay rights	3.52%	3.04%	Remains substantial in every window after a Y2 high.
Career precarity and exit	2.54%	2.47%	Broadly stable across four rolling windows.

All percentages divide posts in included candidate-theme clusters by every filtered post in the same rolling window. They show the portion of the filtered corpus captured by this 27-cluster map, not a complete measure of all management-related content.

Data collection

Public r/antiwork submissions were retrieved through the Arctic Shift archive API in monthly UTC batches from 1 March 2021 through 1 March 2025. The analysis retained unique posts whose title or body matched the established management terms boss, manager, supervisor, or team lead, including plural variants. The resulting corpus is an archive-based sample of public discourse, not a representative sample of employees or an author-level panel. Archive records may differ from content displayed on Reddit after collection.

Analytic workflow

The analysis proceeded as follows. Corpus checks, model-based topic discovery, descriptive prevalence estimates, and robustness evidence are reported separately.

- 1. Freeze and audit the corpus.** Matched the management-related filter to the preprocessed analysis corpus and confirmed 97,254 unique post IDs. A duplicate audit found 1.2% residual duplicate non-placeholder texts; this is reported rather than silently removed.
- 2. Construct the primary text field.** Used post title plus body after handling deleted and placeholder content. A title-only run was retained as a sensitivity analysis, not substituted for the primary representation.
- 3. Discover topics with a current topic-modeling pipeline.** Embedded documents with all-MiniLM-L6-v2, reduced the embedding space with UMAP, clustered with HDBSCAN, and represented topics with BERTopic c-TF-IDF. The full-corpus model produced 199 non-outlier clusters and assigned 38.6% of posts.
- 4. Document the theme map.** Reviewed the largest 30 clusters and retained 27 interpretable clusters in a nine-theme candidate codebook. Generic, deleted, and community-meta clusters were excluded, yielding 18,506 mapped posts (19.0% of the full filtered corpus).
- 5. Estimate descriptive longitudinal patterns.** Calculated each theme's share of every filtered post in four March-to-February rolling windows. The figures in this brief show the main monthly patterns, the full reviewed-theme comparison, and monthly assignment coverage.
- 6. Test stability and state the boundary.** Ran seed-stability checks, title-only sensitivity, and one-to-one top-term alignment checks across rolling-window fits. Sentiment was intentionally not analyzed, and the paper will make no causal or full-conversation prevalence claims.

The private repository includes the analysis scripts, codebook, diagnostics, and aggregate evidence used in this brief. Raw posts and text-level samples remain local.

From collection to estimates

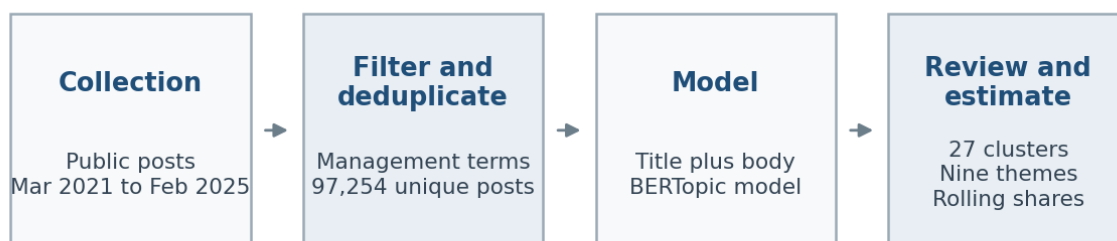
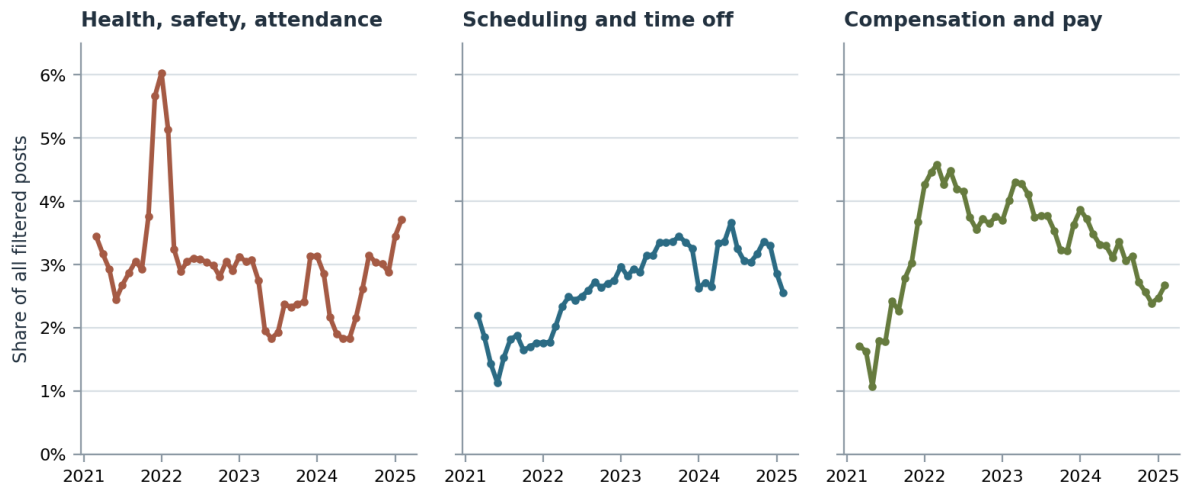


Figure 1. The collection, filtering, modeling, review, and estimation sequence used for the exploratory analysis.

Monthly patterns in the main themes



Three-month rolling shares; denominator is all filtered posts each month. The reviewed theme map covers 19.0% of the corpus.

Figure 2. Three-month rolling prevalence of the themes that anchor the main interpretation. Shares use all filtered posts each month as the denominator.

All reviewed themes by rolling window

Provisional theme prevalence by rolling window

Share of all filtered posts. Exploratory 27-cluster map excludes generic, deleted, and community-meta clusters.

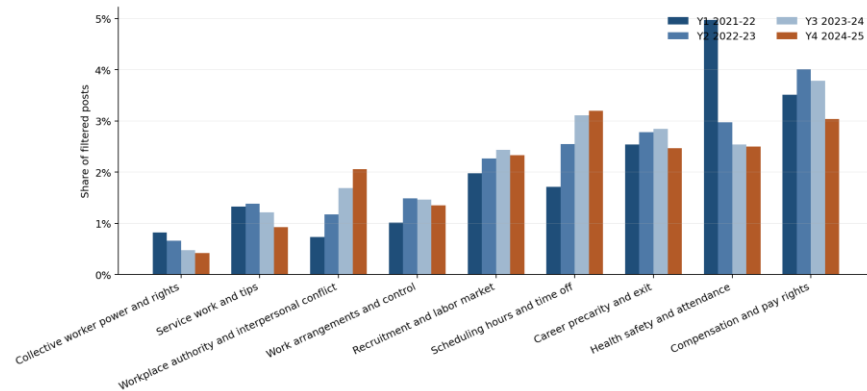


Figure 3. Exploratory prevalence by rolling window. The 27-cluster map excludes generic, deleted, and community-meta clusters.

Analytic approach and scope

Primary analysis

Sentence-transformer embeddings (all-MiniLM-L6-v2), UMAP, HDBSCAN, BERTopic topic representations, and a full-corpus model for longitudinal prevalence. Per-window fits serve as stability diagnostics.

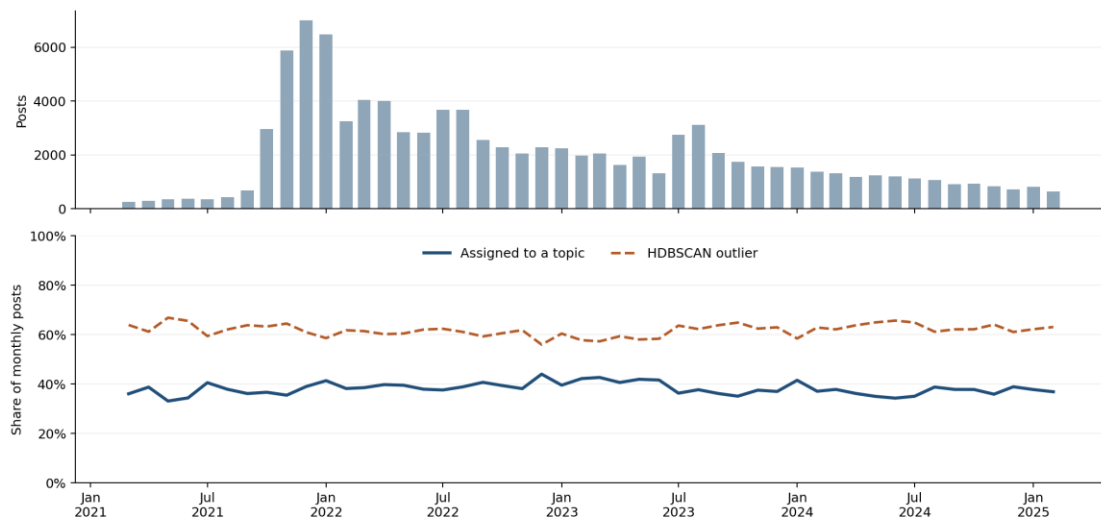
Coverage of estimates

The model assigns 38.6% of posts to non-outlier clusters. The final theme map covers 18,506 posts: 19.0% of all filtered posts and 49.3% of assigned posts.

Assignment coverage

Full-corpus BERTopic assignment coverage

97,254 filtered r/antiwork posts, March 2021 through February 2025



Interpret raw topic trends alongside this changing assignment rate; outliers are not a substantive category.

Figure 4. Assignment coverage varied from 33.1% to 44.0% by month. Outliers are a coverage limitation, not a substantive category.

Robustness checks

Check	Result
Corpus audit	Inputs contain 97,254 matching unique post IDs; the management-term filter re-matches 100% of retained posts.
Seed stability	Across three UMAP seeds on the same 10,000-post stratified sample, joint-inlier ARI was 0.900 to 0.966.
Window alignment	Conservative one-to-one term matching is supporting context, not proof of substantive persistence.
Text sensitivity	Titles-only clustering was stable but materially different from title-plus-body; title-plus-body remains the primary representation.

A larger all-mpnet-base-v2 encoder check was attempted but did not complete within the available CPU bound, so it is deferred and not treated as evidence.

Interpreting the results

Observed patterns

Within the mapped themes, health, safety, and attendance discussion declined over the four windows, while scheduling, hours, and time-off discussion increased. Compensation, career precarity, and recruitment remained visible throughout.

What the results support

These patterns describe changes in the composition of management-related public discussion. They identify domains for follow-up with representative employee data; they do not establish employee sentiment, turnover drivers, or policy effects.

Remaining work

Available for drafting

Results, figures, a theme codebook, coverage diagnostics, seed stability, text-mode sensitivity, and a defined interpretive scope are ready for the manuscript.

Out-of-scope extensions

A broader cluster review, an independent encoder check when compute permits, or a future sentiment study with a validated human reference set can be considered separately. None are required for this paper.

Manuscript revision status

Sections revised

The revised manuscript includes the title and abstract; current-study framing and research question; Methods, Results, Discussion, implications, limitations; figures; and primary BERTopic references.

Before circulation

Correct the denominator language, add the archive-source statement and sampling-frame framing note, and harmonize retained Background language with the paper's exploratory, descriptive scope. Historical theory and prior turnover findings should remain context, not claims tested by this study.

Sampling-frame framing

Intended analytic focus

The management-term filter intentionally concentrates a problem-oriented, likely negatively skewed subset of r/antiwork discussion. The purpose was to surface recurring management concerns that may warrant follow-up.

Boundary for the manuscript

Do not present the results as average employee sentiment or a standalone priority ranking. Frame the themes as candidate areas to examine alongside representative evidence and organizational context.

Materials for drafting

Draft in this order: method and sampling frame; coverage and descriptive results; limitations; then an exploratory discussion. Use the documented theme map and avoid causal, full-conversation, or sentiment claims.

Reproducible code and shareable evidence: <https://github.com/tristowww/antiwork-topic-modeling>

Method records: planning/SCRAPING.md, planning/PREPROCESSING.md, planning/WRITING_HANDOFF.md, planning/ROBUSTNESS.md, and planning/CANDIDATE_THEME_CODEBOOK.csv.

Current status: the analysis outputs are complete. The manuscript needs a focused wording pass before circulation as a descriptive, exploratory topic-prevalence study.

Appendix: Reviewed cluster map

The table below makes the review trail visible. It lists all 30 largest clusters examined for the candidate-theme map, their representative terms, the assigned candidate theme and subtheme, and the inclusion decision.

Reviewed clusters 1-15

ID	Representative terms	Candidate theme and subtheme	Decision
0	interview hiring interviews recruiter applied resume	Recruitment and labor market Hiring and interviews	Include high confidence
1	bad good hate conversation like worst	Other Generic/deleted manager posts	Exclude high confidence
2	raise raises increase salary inflation pay	Compensation and pay rights Raises and salary	Include high confidence
3	rehearsal good team bad employees people	Other Generic manager-role discussion	Exclude medium confidence
4	covid positive tested test sick symptoms	Health safety and attendance COVID and workplace illness	Include high confidence
5	check paycheck paid pay deposit payroll	Compensation and pay rights Payroll and withheld wages	Include high confidence
6	remote office home hybrid wfh remotely	Work arrangements and control Remote hybrid and return-to-office	Include high confidence
7	coworker like things said just doesn	Workplace authority and interpersonal conflict Interpersonal mistreatment and role overload	Include medium confidence
8	life school want career college money	Career precarity and exit Economic strain and career pressure	Include medium confidence
9	notice weeks period week giving resignation	Career precarity and exit Notice periods and resignation	Include high confidence
10	schedule availability scheduled saturday sunday days	Scheduling hours and time off Scheduling and availability	Include high confidence
11	pto vacation use approved days request	Scheduling hours and time off PTO and unpaid time off	Include high confidence
12	tips tip tipping restaurant pool cash	Service work and tips Tipping and tip withholding	Include medium confidence
13	vacation request trip days approved time	Scheduling hours and time off Vacation denial	Include high confidence
14	union unions strike unionize workers unionizing	Collective worker power and rights Unionization and strikes	Include high confidence

Reviewed clusters 16-30

ID	Representative terms	Candidate theme and subtheme	Decision
15	restaurant kitchen chef cook food bar	Service work and tips Hospitality and restaurant conditions	Include high confidence
16	break breaks lunch minute 30 minutes	Scheduling hours and time off Meal and rest breaks	Include high confidence
17	review reviews performance glassdoor feedback rating	Workplace authority and interpersonal conflict Performance reviews feedback and employer reputation	Include medium confidence
18	quit quitting resign leaving tell today	Career precarity and exit Quitting decisions	Include high confidence
19	funeral bereavement died mom passed family	Health safety and attendance Bereavement and compassionate leave	Include high confidence
20	promotion promoted position role team level	Career precarity and exit Promotion and advancement	Include high confidence
21	mental health autistic disability autism disabled	Health safety and attendance Mental health and disability	Include high confidence
22	wages discuss discussing illegal nlrb pay	Compensation and pay rights Pay transparency and wage discussion rights	Include high confidence
23	pregnant pregnancy baby maternity leave paternity	Health safety and attendance Pregnancy and parental leave	Include high confidence
24	snow weather storm roads hurricane ice	Health safety and attendance Weather and commute safety	Include high confidence
25	sick calling called come days feeling	Health safety and attendance Sick leave and attendance pressure	Include high confidence
26	fired firing got adverse threatened activity	Career precarity and exit Firing retaliation and termination	Include medium confidence
27	overtime exempt ot hours salaried 40	Compensation and pay rights Overtime and exempt status	Include high confidence
28	phone phones personal cell app number	Work arrangements and control Personal-phone and off-hours boundaries	Include high confidence
29	antiwork sub anti movement subreddit society	Other Community meta-discussion	Exclude high confidence